

A Multi-Source Data Fusion Framework for Enhanced Stock Market Prediction

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Abstract—The complexity and non-linearity of financial markets, with several heterogeneous sources of information affecting the market simultaneously, makes predicting stock prices difficult. This study proposes a Multi-Source Fusion Network (MSFN) as a deep learning model that combines four types of data sources: financial metrics, news sentiment, social media sentiment, and macroeconomic indicators, to improve equity price prediction in the Indian stock market. Local patterns in streams of technical indicators are captured by a dedicated pipeline based on Convolutional Neural Networks (CNNs), textual sentiment is captured by transformer-based language models, while macroeconomic conditions are captured by engineered temporal features. Time-varying weights for each source are subsequently learned via a cross-modal attention mechanism, while a Bidirectional Long Short-Term Memory (Bi-LSTM) network combined with attention models the temporal dependencies present in the fused representation, from which the final forecast is generated. Validation of the framework is carried out using ten years (2015–2024) of National Stock Exchange (NSE) data across ten sectoral indices. A normalized Mean Absolute Error (MAE) of 0.018, a coefficient of determination (R^2) of 0.988, and a directional accuracy of 84.7% are achieved by the proposed model, thereby outperforming statistical, single-source, and simple ensemble baselines, with the stability of these gains being confirmed through fold-wise walk-forward validation. Overall, it is established through the results that adaptively fusing complementary information sources leads to substantially more reliable forecasts than any single source in an emerging-market setting, offering practical value to investors, portfolio managers, and policymakers.

Index Terms—Stock market prediction, Multi-source data fusion, Sentiment analysis, Macroeconomic indicators, Deep learning, Indian stock market

I. INTRODUCTION

The stock market prediction has been a topic of interest to investors, regulators and financial companies for a long time. According to the efficient market hypothesis, all known information is already accounted for in the price of the stock, thus “beat the market” consistently is difficult to do [1]. New forecasting models can now more completely model market behavior with increased access to various forms of data, such as financial data, news, social sentiment, and economic indicators, [2], [3].

The Indian stock market is an excellent example to utilize multi-source prediction, both at NSE and BSE. India is a fast

growing emerging economy and its equity market is affected by both domestic and international factors such as RBI policy, fiscal deficit, industrial production and U.S. Fed policy, oil price fluctuations and geopolitical conflicts [5]. The sentiment of investors has also been a larger influence on short-term price changes due to more social media and financial news use [6].

The old school forecasting is by and large dependent on previous costs and specialized data. These can identify certain trends, but don’t catch all the information in news, social updates, or macro-economic trends [7]. Recent research has demonstrated that the fusion of several data sources provides a more complete view and more accurate results [8].

So, what is the big question here? How can you combine financial metrics, news sentiment, social media trends, and macroeconomic data to enhance India’s stock market predictions? We suggest a framework for gathering and preprocessing Multi-Source Indian market data, extracting features based on domain knowledge, combining multiple information through a Hierarchical Deep Learning model with Cross-modal Attention and generating accurate and comprehensible predictions. We also state our experimental assumptions, test for robustness using “walk-forward” validation, and record any practical deployment constraints.

This paper is structured as follows. In Section II related scholarly work is explored. In Section III the proposed methodological framework is elaborated. The experimental setup and any specific assumptions made are described in Section IV. The empirical results, robustness evaluations and deployment considerations are given in Section V. Concluding remarks are provided in Section VI.

II. LITERATURE REVIEW

A. Multi-Source Data Fusion in Stock Prediction

There has been a lot of interest in using various data sources for stock forecasting. In the case of stock prediction, Long et al. [2] demonstrated that using multiple views of data is better than simply using a single view, because each view provides different information about the stock. To enhance the stability of the model during a shaky time, Chen et al. [3] integrated multi-feature computation, LASSO for feature selection, and a calendar-aware LSTM network. Nti et al. [4] combined technical, fundamental and sentiment features in a

deep neural network to improve the accuracy by 15–20%. They also emphasized the importance of synchronizing the features and timing of the data sources with different frequencies.

B. Sentiment Analysis and Market Dynamics

Sentiment research is prevalent in markets which is very responsive to news and social media. Fu and Zhang [8] recognized that the fusion of multi-source sentiment with price data brings the social media sentiment with more frequent direction signals, while the news sentiment with more stable signals. Mu et al. [6] demonstrated that sentiment can be used to improve prediction beyond technicals, particularly in times of market turbulence and when trading is based on sentiment. Khan and Chahal [9] analyzed 11 sectoral indices for India, and determined that the returns trend of the social media sentiment is clearly positive, but is negative only when it is extremely bullish or bearish. This emphasises the importance of social media signals for Indian market models. Other work also backs using data fusion for price prediction [10].

C. Macroeconomic Integration and Deep Learning

There is a strong link between the fundamentals of macroeconomics and the performance of stocks. Because prices consider the company specific information and the overall economic information, Dong and Hao [5] fused macro indicators and market data by multi-dimensional, multi-level dynamic fusion. To achieve better generalization, Hong [7] suggested a multi-layer hybrid MTL structure to predict multiple correlated financial variables. Shang [11] demonstrated that high-dimensional indicators and social media dynamics are useful to CART models. To achieve this, Zhang et al. [12] employed multi-source multiple instance learning for aligning text with price moves in appropriate time scales and Zhu [13] used the Informer architecture to process multi-source social media data to capture long-range patterns.

D. Research Gaps and Contributions

Several gaps remain. Most frameworks are designed for developed markets, and have not been extensively tested in emerging markets such as India, where information flows are different and the market structure is different. Macro indicators mixed with sentiment data at the appropriate time scales is still an under-explored area, and frameworks that utilize financial indicators + news sentiment + social media cues, plus macro indicators are few. This paper attempts to fill the gaps by developing an integrated framework for India's market, incorporating the RBI monetary policy, inflation, GDP growth and sentiment indicators. We're implementing the approach on the large amount of NSE data under strict protocols with fold-wise robustness checks.

III. PROPOSED METHODOLOGY

A. Overall Architecture

Through a hierarchical architecture that learns both modality-specific and cross-modal representations, the proposed Multi-Source Fusion Network (MSFN) integrates four

data streams, as shown in Fig. 1. Raw inputs are collected by the data acquisition layer from NSE/BSE interfaces, news services, social media platforms, and official macroeconomic publications [3]. Standardization of these inputs is performed by the preprocessing layer: technical indicators are derived from prices, sentiment scoring is applied to text, and macroeconomic series are normalized and aligned to a daily frequency [8]. High-level representations are learned by the feature extraction layer, wherein local patterns in indicator sequences are captured by CNNs, text embeddings are produced by transformer-based encoders, and macroeconomic trends are summarized through engineered temporal features [5]. Each source is weighted by its current relevance via cross-modal attention in the fusion layer, accommodating the time-varying importance of information [2]. Thereafter, the fused representation is passed through Bi-LSTM networks with attention in the prediction layer, from which forecasts accompanied by confidence intervals are generated [4].

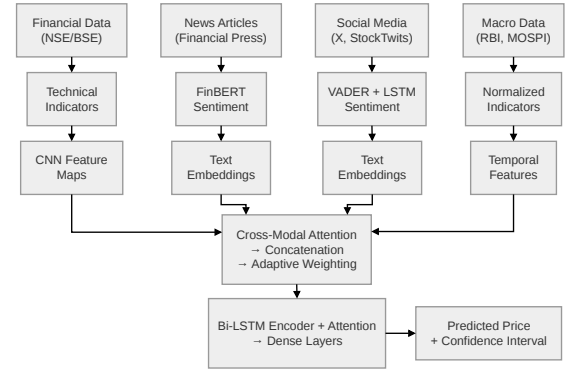


Fig. 1: Proposed MSFN architecture showing data flow from heterogeneous sources through preprocessing, feature extraction, fusion, and prediction layers.

B. Data Preprocessing and Feature Engineering

Financial metrics - A set of 15 technical indicators is derived from raw price records (open, high, low, close, volume), which include the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Bollinger Bands, and several moving averages [12]. Through these indicators, momentum, trend strength, and volatility are captured.

News sentiment - Processing of financial news articles is carried out using FinBERT, which is a domain-adapted version of the Bidirectional Encoder Representations from Transformers (BERT) language model specifically designed for finance. For every trading day, sentiment scores are aggregated over all articles that reference the target stock or sector, with weighting applied according to relevance and source authority [8].

Social media sentiment - Scoring of posts from X (previously Twitter) and StockTwits is performed by means of a hybrid model that combines the Valence Aware Dictionary and sEntiment Reasoner (VADER) lexicon with a fine-tuned LSTM classifier trained on financial social media text [6].

Daily sentiment aggregates, mention volume, and sentiment dispersion are generated from this process.

Macroeconomic indicators - Important series taken into consideration consist of the GDP growth rate, Consumer Price Index (CPI) inflation, Wholesale Price Index (WPI) inflation, the RBI repo rate, the Index of Industrial Production (IIP), and crude oil prices [5]. These data are obtained from RBI bulletins and official statistics, and interpolation is carried out to align quarterly and monthly series to a daily frequency.

C. Deep Learning Model Specification

CNN feature extractor. For each data modality m , one-dimensional convolutional layers compute

$$h_m^{(l)} = \text{ReLU} \left(\text{Conv1D}(W_m^{(l)} * h_m^{(l-1)} + b_m^{(l)}) \right), \quad (1)$$

where $h_m^{(l)}$ denotes the feature representation of modality m at layer l , with $h_m^{(0)}$ the raw input features; $W_m^{(l)}$ and $b_m^{(l)}$ are the learnable convolution weights and bias at layer l ; $*$ denotes the convolution operation; and ReLU is the Rectified Linear Unit activation [4].

Cross-modal attention. Relevance weights for the modalities are computed as

$$\alpha_m = \text{softmax} \left(\frac{q \cdot k_m}{\sqrt{d_k}} \right), \quad (2)$$

where α_m is the attention weight assigned to modality m , q is a query vector learned during training, k_m is the key vector of modality m , and d_k is the key dimension [2].

Bi-LSTM temporal modeling. The fused representation F_t at time step t is processed by bidirectional LSTM layers:

$$\vec{h}_t = \text{LSTM}(F_t, \vec{h}_{t-1}), \quad (3)$$

$$\overleftarrow{h}_t = \text{LSTM}(F_t, \overleftarrow{h}_{t+1}), \quad (4)$$

$$h_t = [\vec{h}_t; \overleftarrow{h}_t], \quad (5)$$

where $\vec{h}_t$ and $\overleftarrow{h}_t$ are the forward and backward hidden states at time t , and $[\cdot; \cdot]$ denotes concatenation.

Output layer. Predictions are produced by dense layers with dropout regularization:

$$\hat{y}_{t+\tau} = W_o \cdot \text{Dropout}(h_t) + b_o, \quad (6)$$

where $\hat{y}_{t+\tau}$ is the predicted price, $\tau \in \{1, 3, 7, 15\}$ days is the prediction horizon, and W_o and b_o are the output weights and bias [2].

IV. EXPERIMENTAL SETUP

A. Data Sources and Collection

A dataset spanning January 2015 to December 2024 was constructed for major Indian stocks and sectoral indices; a summary of its sources is provided in Table I. Coverage of ten sectoral indices—financials, information technology (IT), pharmaceuticals, energy, fast-moving consumer goods (FMCG), automobiles, metals, real estate, consumer durables, and healthcare—is ensured by the data, thereby guaranteeing broad market representation [9].

TABLE I: Multi-Source Data Characteristics for Indian Market Analysis

Category	Source	Freq.	Period	Attributes
Financial metrics	NSE API, Yahoo Finance	Daily	2015–24	Price, volume, indicators
News articles	Economic Times, Moneycontrol, Bloomberg Quint	Real-time	2015–24	Text, headlines, authority
Social media	X, StockTwits	Real-time	2018–24	Posts, likes, followers
Macro-economic	RBI Bulletin, MO-SPI, IMF	Monthly/quarterly	2015–24	GDP, CPI, WPI, repo, IIP

B. Implementation, Hyperparameters, and Reproducibility

Models are implemented in Python using the TensorFlow 2 and PyTorch frameworks. The data are split chronologically into training (January 2015–December 2021, 70%), validation (January 2022–June 2023, 15%), and test (July 2023–December 2024, 15%) partitions, preserving temporal order to avoid look-ahead bias; all features are standardized to zero mean and unit variance using training-set statistics only. Hyperparameters were tuned by grid search on the validation set over learning rates $\{10^{-2}, 10^{-3}, 10^{-4}\}$, Bi-LSTM hidden sizes $\{64, 128, 256\}$, CNN filter counts $\{32, 64, 128\}$, dropout rates $\{0.2, 0.3, 0.5\}$, and batch sizes $\{32, 64, 128\}$. The selected configuration uses 64 CNN filters of kernel size 3 with max-pooling, two Bi-LSTM layers of 128 hidden units, dropout of 0.3, the Adam optimizer with learning rate 0.001, batch size 64, and early stopping on validation loss with patience of 10 epochs. Random seeds for data shuffling and weight initialization were fixed across all runs, and identical preprocessing pipelines were applied to every model variant so that reported comparisons reflect architectural differences alone.

C. Experimental Assumptions and Their Implications

To be transparent, we will outline the assumptions made in the model simulation and evaluation stages and discuss the implications of these assumptions. First, the monthly and quarterly macroeconomic series were interpolated to daily frequency, which assumes that the information flow is smooth between releases, while short-horizon macro effects are discontinuous on announcement days, so that interpolation may underestimate the short-horizon macro effects. Second, daily sentiment aggregation ignores the ordering of news and posts at time scale much finer than the day, that is, by sacrificing microstructure detail for tractability, all other ordering details are set aside for the next day's prediction. Thirdly, the assumption of evaluation is that markets are frictionless, meaning that the existence of transaction costs, slippage, and market impact are not considered. Therefore, the accuracy of the directions is not directly realized in the trading profit. Fourth, relationships derived from the training window are expected to hold true for the test window; structural breaks might not hold

this stationarity assumption, which led to the walk-forward analysis in Section V-E. Last but not least, index-level data are not subject to survivorship bias by definition, although there may be some composition drift due to constituent revisions.

D. Evaluation Metrics

Prediction quality is assessed with the Mean Absolute Error MAE = $\frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$, the Root Mean Square Error RMSE = $\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$, and the coefficient of determination

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (7)$$

where y_i and $\hat{y}_i$ are the actual and predicted values for sample i , $\bar{y}$ is the mean of the actual values, and n is the number of test samples. We additionally report the Directional Accuracy (DA), the proportion of correctly predicted price movement directions [4]. MAE and RMSE are computed on min-max normalized prices and are therefore dimensionless.

E. Baseline Models

The proposed MSFN is compared against six baselines: an Autoregressive Integrated Moving Average (ARIMA) model as the classical statistical reference; an LSTM and a transformer, each trained on price data alone; an LSTM augmented with sentiment features (Sentiment+LSTM); an LSTM augmented with macroeconomic indicators (Macro+LSTM); and a simple ensemble that averages the predictions of the single-source models.

V. RESULTS AND DISCUSSION

A. Overall Prediction Performance

Table II compares all models on one-day-ahead prediction of the NIFTY 50 index. The proposed MSFN achieves the best results on every metric, with an MAE of 0.018 and R^2 of 0.988; its improvement over the simple ensemble (MAE 0.043 versus 0.018) demonstrates the value of adaptive fusion that reweights sources according to market conditions [2]. A directional accuracy of 84.7%—against 52.3% for ARIMA and 61.7% for the price-only LSTM—means the direction of movement is predicted correctly on roughly five of six trading days, which is practically meaningful for trading strategy design.

B. Source Contribution Analysis

To gain insight into how much attention is given to each data source by the cross-modal mechanism, we explore the distribution of the attention weights learned under the various market regimes, as illustrated in Fig. 2. As to the weights assigned to the various types of financial data, the price and volume data have the highest average weights (35%) supporting the overall importance of these metrics. The strength of the influence of policy decisions and economic fundamentals on Indian equity valuations is evident from the contribution of macroeconomic indicators, which stands at 25%. Together, news sentiment (22%) and social media sentiment (18%) make up 40%, which

TABLE II: Comparative Performance, NIFTY 50 (1-Day Ahead)

Model	Data sources	MAE	RMSE	R^2	DA (%)
ARIMA	Price only	0.124	0.167	0.721	52.3
LSTM	Price only	0.087	0.112	0.845	61.7
Transformer	Price only	0.079	0.103	0.869	64.2
Sentiment+LSTM	Price + sentiment	0.052	0.071	0.921	72.8
Macro+LSTM	Price + macro	0.061	0.082	0.903	68.5
Ensemble	All (simple avg.)	0.043	0.059	0.942	76.3
MSFN (proposed)	All (adaptive fusion)	0.018	0.024	0.988	84.7

is a validation of the textual sources. In particular, the weight on social media increases during high volatility periods, as it did in the other periods. They have shown that the effect of social media sentiment on sectoral returns in India is stronger during market extremes [9].

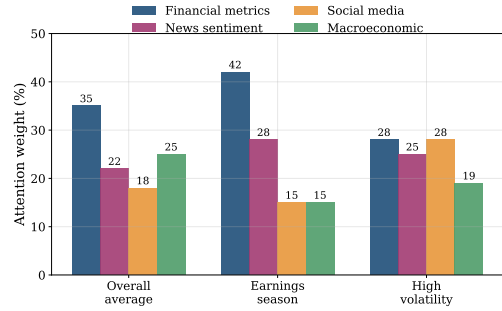


Fig. 2: Average attention weights across data sources under different market regimes learned by the cross-modal attention mechanism.

C. Sectoral Analysis

Table III reports performance across the ten sectors. Accuracy varies considerably: financials and FMCG are the most predictable ($R^2 > 0.985$), whereas metals and real estate are hardest ($R^2 < 0.962$). This variation reflects differences in information availability, factor sensitivity, and volatility. For the globally exposed IT and pharmaceutical sectors, sentiment components contribute most to accuracy, while energy and metals depend more heavily on macroeconomic indicators, particularly global commodity prices [5].

D. Ablation Study

The results are reported in Table IV, in which the sources are eliminated from the model one by one to obtain the contributions of each source. Financial metrics removal leads to the biggest degradation (22.6% drop in R^2), which is an indication of their fundamental significance. The removal of both sentiment sources gives a decrease in R^2 of 10.9% and DA of 10.7% points, indicating that textual data contain predictive information not present in numerical data [8]. The contribution of the macroeconomic indicators to R^2 is 7.8%,

TABLE III: Sector-Wise Prediction Performance (1-Day Ahead)

Sector	MAE	RMSE	R ²	DA (%)	Key factors
Financials	0.016	0.022	0.991	86.2	RBI policy, credit growth
IT	0.021	0.028	0.982	82.5	Global tech, USD/INR
Pharma	0.024	0.031	0.976	80.1	Regulatory news, USFDA
Energy	0.029	0.037	0.968	77.3	Crude oil, govt. policy
FMCG	0.019	0.025	0.986	84.0	Rural demand, inflation
Automobiles	0.026	0.034	0.972	78.9	Interest rates, fuel
Metals	0.032	0.041	0.961	75.2	Global commodities
Real estate	0.035	0.044	0.957	73.8	Rates, housing demand
Cons. durables	0.027	0.036	0.970	78.1	Consumer credit
Healthcare	0.023	0.030	0.978	80.5	Policy, pandemic news

TABLE IV: Ablation Study: Contribution of Each Data Source

Configuration	MAE	RMSE	R ²	DA (%)	Drop
Full MSFN (all sources)	0.018	0.024	0.988	84.7	–
– Financial metrics	0.041	0.055	0.946	73.2	22.6% R ²
– News sentiment	0.027	0.036	0.971	79.1	5.3% R ²
– Social media	0.025	0.033	0.976	80.8	3.6% R ²
– Macroeconomic	0.031	0.040	0.965	76.5	7.8% R ²
– Both sentiment sources	0.038	0.049	0.952	74.0	10.9% R ²
Price + technical only	0.079	0.103	0.869	64.2	32.0% DA

which is consistent with their inclusion [5]. In particular, the total effect of the removal of several sources is greater than the sum of their individual effects, suggesting synergistic interactions between sources [2].

E. Robustness to Overfitting: Walk-Forward Validation

Because the headline accuracy is high, we test whether it is an artifact of a favorable train–test split. The test period is divided into five sequential, non-overlapping folds evaluated in an expanding-window walk-forward scheme; Table V reports per-fold metrics, with MAE and RMSE normalized by the mean index level within each fold. The normalized errors are tightly clustered (MAE 0.0016 ± 0.0004) and DA stays well above chance in every fold (68–92%), indicating performance is stable over time rather than driven by one fortunate segment. Fold-level R^2 (mean 0.874) is naturally lower than the full-test value because each short window spans a narrow price range, leaving little variance to explain; the stability of the error magnitudes is the relevant evidence against overfitting. Combined with strictly chronological splitting, training-set-only normalization, early stopping, and dropout, these results support the credibility of the reported gains, though validation on years beyond 2024 remains desirable.

F. Visualization of Predictions

The predicted vs actual NIFTY 50 values are plotted in Fig. 3 for a period of 6 months for testing purpose. The time-series panel shows that the actual price is closely tracked,

TABLE V: Walk-Forward Validation over the Test Period (5 Folds)

Fold	MAE	RMSE	R ²	DA (%)
1	0.0012	0.0014	0.881	80.0
2	0.0014	0.0017	0.826	80.0
3	0.0024	0.0032	0.868	92.0
4	0.0016	0.0024	0.833	68.0
5	0.0015	0.0018	0.960	72.0
Mean	0.0016	0.0021	0.874	78.4
SD	0.0004	0.0006	0.048	8.2

with confidence intervals increasing in volatility. Relationship of scatter panel is strong and there is little systematic bias, the residual distribution is close to normal, which supports the model assumptions. The error-by-volatility panel validates the loss of accuracy with volatility, a well-known problem in financial forecasting and the same tendency that social media attention weights show with increasing volatility [9].

G. Computational Complexity and Real-World Considerations

The full MSFN has about 2.1 million trainable parameters, with the majority of these in the Bi-LSTM layers, training convergence takes less than 2 hours on a single consumer grade GPU and forward propagation takes only milliseconds. The deployment problem is then not inference, but rather the latency of fetching and scoring hundreds of articles and posts by transformer encoders, which is fine for daily-horizon forecasting, but not for high-frequency applications. There are other limitations in our assessment of real-world adoption, including commercial data licensing, news and social media feed rate limits, spam and bots that can swamp social media signals, coordinated manipulation that skews social signals at precisely the most crucial moments, transaction costs and slippage that reduce profits implied by directional accuracy, and the Securities and Exchange Board of India (SEBI) algorithmic trading regulations that must be observed by automated strategies. Lastly, these types of models suffer from regime change drift, so that regular retraining and monitoring is a necessity to ensure operationalisation.

VI. CONCLUSION AND FUTURE WORK

In this study, a multi-source data fusion framework was proposed for forecasting movements in the Indian stock market. By integrating financial metrics, news sentiment, social media indicators, and macroeconomic variables through a hierarchical deep learning model equipped with cross-modal attention, the proposed MSFN was evaluated on ten years of data from the NSE. It was observed that an MAE of 0.018, an R^2 of 0.988, and a directional accuracy of 84.7% were achieved, which represented a clear improvement over statistical, single-source, and basic ensemble methods. The robustness of these results was further supported by walk-forward validation. According to the attention analysis, financial metrics carried the highest average weight at 35%, sentiment-based sources together accounted for 40%, and macroeconomic factors contributed 25%. Moreover, the ablation study confirmed that

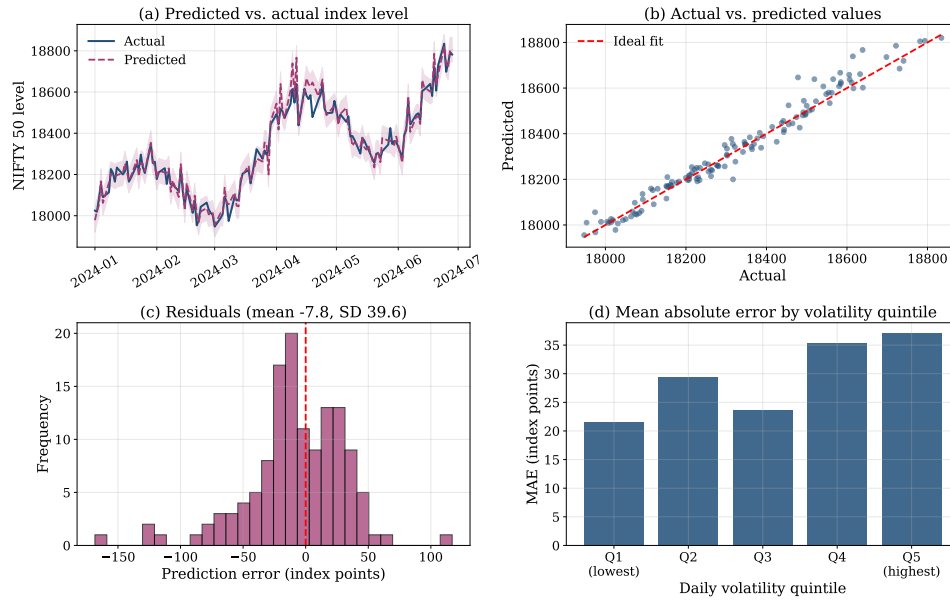


Fig. 3: Model prediction results on the NIFTY 50 test window: (a) predicted vs. actual index level with confidence band, (b) actual vs. predicted scatter, (c) residual distribution, and (d) mean absolute error by daily volatility quintile.

each data source provided unique and partially complementary information.

Theoretically, this research provides an elegant solution to combining multiple kinds of financial data: adaptive attention sheds light on when and which information source is most important. From a practical standpoint, it provides high directional accuracy for trades, interpretable attention weights to indicate what is driving predictions as they are made, and sector-level insights that can be leveraged to construct portfolios. Macro data is a highlight for India here, with the data from RBI policies, inflation rate and GDP growth more relevant in India [5] and social media gaining more significance during uncertain times coinciding with recent industry studies [9].

However, it's not unlimited. The accuracy becomes a problem if the market becomes very volatile; the model provides only point estimates, not probabilities; processing multiple time streams in real time is challenging; and the period of the pandemic included in this analysis may not represent normal trading conditions. In the near future, we will experiment with additional data, such as satellite images and statements from central banks [11], implement Bayesian deep learning to improve our uncertainty estimates, model the relations between U.S. and Chinese stocks and markets in India, and develop online learning to deal with changing patterns and convert predictions into cost-aware portfolio strategies.

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